

National Taiwan University
Department of Economics
ECON 4037: Game Theory
Homework 1

Due 2:20 PM on Wednesday, September 23, 2026
You may submit online or hand in a handwritten copy.

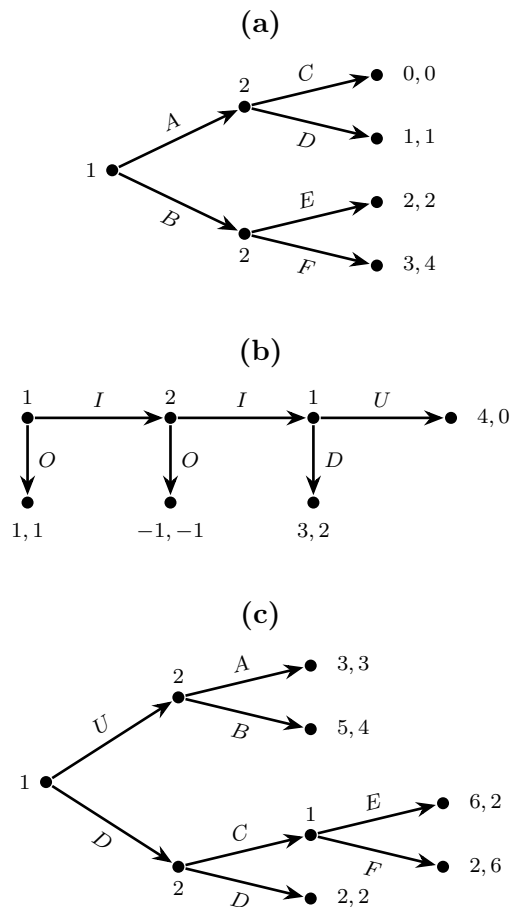
Show your work. You may discuss these problems with classmates or use generative AI tools while working on them, but you must write your own answers and be able to explain and defend everything you submit (see the syllabus's Course Requirements).

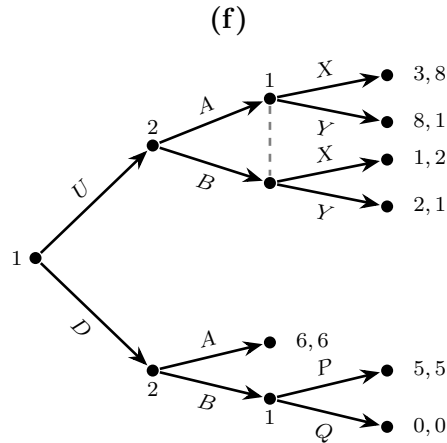
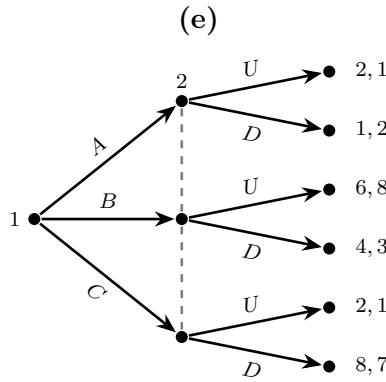
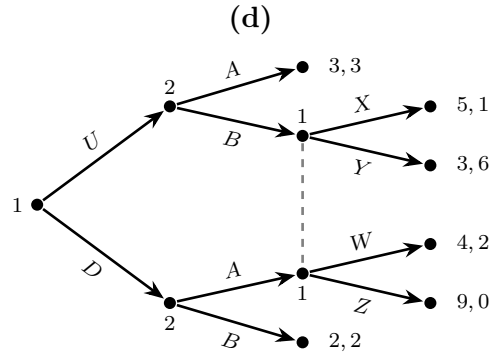
- There are three people: Amy, Bart, and Chris. Amy and Bart have hats. They are arranged in a room so that Bart can see everything Amy does, and Chris can see everything Bart does, but Chris cannot see what Amy does, and no one can see anything else.

First, Amy chooses either to put her hat on her head (H) or on the floor (F). After observing Amy's move, Bart chooses between putting his hat on his head or on the floor. If Bart puts his hat on his head, the game ends and everyone gets a payoff of 0. If Bart puts his hat on the floor, Chris must guess whether Amy's hat is on her head, by saying either "yes" or "no," which ends the game. If Chris guesses correctly, he gets a payoff of 1 and Amy gets -1 ; if he guesses incorrectly, these payoffs are reversed. Bart's payoff is always 0.

Represent this game in the extensive form (draw the game tree), labeling every information set.

- Draw the normal-form matrix of each of the following extensive-form games.





3. Consider a version of the *Cournot duopoly game*. Two firms, 1 and 2, compete in a homogeneous-goods market, simultaneously and independently selecting quantities $q_1, q_2 \geq 0$ to produce. The market price is $p = 2 - q_1 - q_2$. Production is costless, and each firm's payoff is its profit, $u_i = pq_i$.

Describe the normal form of this game: write down each firm's strategy space, and write each firm's payoff as a function of the strategy profile.

4. Evaluate the following payoffs for the game given by the normal form pictured here. A mixed strategy for player 1 is $\sigma_1 \in \Delta\{U, M, D\}$, where $\sigma_1(U)$ is the probability that player 1 plays U , and so forth; we write σ_1 as $(\sigma_1(U), \sigma_1(M), \sigma_1(D))$, and similarly for player 2.

	L	C	R
U	10, 0	0, 10	3, 3
M	2, 10	10, 2	6, 4
D	3, 3	4, 6	6, 6

- (a) $u_1(U, C)$
 - (b) $u_2(M, R)$
 - (c) $u_2(D, C)$
 - (d) $u_1(\sigma_1, C)$ for $\sigma_1 = (1/3, 2/3, 0)$
 - (e) $u_1(\sigma_1, R)$ for $\sigma_1 = (1/4, 1/2, 1/4)$
 - (f) $u_1(\sigma_1, L)$ for $\sigma_1 = (0, 1, 0)$
 - (g) $u_2(\sigma_1, R)$ for $\sigma_1 = (1/3, 2/3, 0)$
 - (h) $u_2(\sigma_1, \sigma_2)$ for $\sigma_1 = (1/2, 1/2, 0)$ and $\sigma_2 = (1/4, 1/4, 1/2)$
5. In lecture, we discussed why a strategy must specify what a player would do at an information set, even one that the player's own earlier choices make impossible to reach (for example, Katzenberg's plan for node e even when he plans to Stay at node a).

In two or three sentences, explain in your own words why this is necessary, and give one example of your own, different from the ones used in lecture, of a real decision that has this same "plan for a contingency you don't expect to face" structure.